

SHAHZAD

Full Stack Developer | AI / ML Engineer

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ABOUT ME

Final-year Computer Science student at NED University of Engineering & Technology with hands-on experience engineering full-stack web applications, AI/ML pipelines, and real-time data systems. Proficient across the complete development lifecycle from architecting REST APIs and MongoDB schemas to deploying React frontends and integrating computer vision models. Seeking software engineering, backend, or full-stack roles where strong fundamentals and production-oriented project experience drive immediate impact.

EXPERIENCE

Frontend Development Intern

Oct 2025 – Nov 2025

CoreTech Innovations · Remote

- Engineered 8 reusable React.js components for a responsive portfolio-style dashboard, accelerating feature delivery and ensuring consistent UI patterns across 5 application sections.
- Optimized frontend architecture for component reusability and separation of concerns, reducing redundant code and improving maintainability.
- Deployed production build to Netlify; collaborated asynchronously in a remote-first team to resolve UI bugs and ship features on schedule.

Simulation & Modelling Intern

Oct 2025 – Dec 2025

EVsim · Remote

- Developed and validated 20+ electric motor simulation scenarios using MATLAB and Xcos, analyzing performance metrics including efficiency, torque, and system behavior under variable load conditions.
- Produced structured technical documentation and validation reports, creating a reusable knowledge base for future simulation iterations.
- Applied systematic debugging and root-cause analysis to resolve model discrepancies and ensure accuracy of simulation outputs.

PROJECTS

Real-Time Vehicle Anomaly Detection System (Final Year Project)

2025 – 2026

Tech Stack: Python, YOLOv11s, OpenVINO, OpenCV, ByteTrack, EasyOCR, Streamlit

- Architected a real-time traffic surveillance pipeline processing live video streams to detect speeding, wrong-way driving, lane violations, and stalled vehicles using YOLOv11s with OpenVINO acceleration.
- Integrated ByteTrack multi-object tracking and asynchronous OCR (ThreadPoolExecutor) for license plate extraction, achieving 43% throughput improvement over synchronous baseline.
- Designed a Streamlit-based monitoring dashboard with live detection overlays, alert visualization, and exportable incident logs for operator review.
- Processed 500+ vehicle instances in validation testing; automated alert mechanism reduced simulated incident response time by 30%.

Smart Campus Portal

2025

Tech Stack: React.js, Node.js, Express.js, MongoDB, Flask, Power BI, JWT

- Architected a full-stack campus management platform with role-based access control (RBAC) for students, faculty, and administrators, managing 1,000+ simulated records across student, course, and analytics collections.
- Built RESTful APIs with Express.js and integrated an AI-powered course recommendation engine and NLP chatbot to enhance academic engagement.
- Implemented real-time Power BI dashboards surfacing performance metrics, skill gaps, and course completion trends for administrative decision-making.

Career Optimization Data Warehouse

2025

Tech Stack: Python, Scikit-learn, Pandas, NumPy, Power BI, SQL

- Designed and implemented a centralized data warehouse aggregating student academic and career data from multiple heterogeneous sources for unified analysis.
- Built and tuned machine learning classification models to generate career pathway recommendations, improving recommendation accuracy by 25% over the rule-based baseline.
- Developed interactive Power BI dashboards visualizing career trends, skill gap analyses, and cohort performance for academic advisors.

Automated Emergency Call Transcript Classifier

2024

Tech Stack: Python, NLP, Scikit-learn, NLTK, Pandas

- Developed an NLP classification system to automatically categorize emergency call transcripts by priority level, applying TF-IDF feature extraction and ensemble classifiers.
- Automated triage workflows end-to-end, eliminating manual categorization effort and improving response prioritization accuracy for downstream dispatch systems.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), Python, SQL, C/C++

Frontend: React.js, HTML5, CSS3, Tailwind CSS, Responsive Web Design, Axios

Backend: Node.js, Express.js, Flask, REST API Design, JWT Authentication

Databases: MongoDB, PostgreSQL, SQL Server, Database Schema Design

AI / ML & CV: TensorFlow, Scikit-learn, YOLO (Ultralytics), OpenCV, OpenVINO, ByteTrack, EasyOCR, Pandas, NumPy

DevOps & Tools: Docker, Git, GitHub, Postman, Jupyter Notebook, Netlify, Vercel, Power BI

Concepts: OOP, Data Structures & Algorithms, SDLC, RBAC, Microservices, Computer Vision, NLP

EDUCATION

Bachelor of Science in Computer Science

2022 – Sep 2026 (Expected)

NED University of Engineering & Technology, Karachi

Relevant Coursework: Data Structures & Algorithms · Database Systems · Software Engineering · Artificial Intelligence · Machine Learning · Computer Networks · Operating Systems

CERTIFICATIONS

- Meta Front-End Developer Professional Certificate — Meta / Coursera (2025)
- Full-Stack Web Development with React — HKUST / Coursera (2025)
- Machine Learning with TensorFlow on Google Cloud — Google Cloud / Coursera (2025)

AWARDS & RECOGNITION

- Full Stack Development Recognition — NED CS Project Showcase (2025)
- Machine Learning Project Merit Award — NED Tech Week (2024)